نـزرع للغـد
ICBA
AGRICULTURE FOR TOMORROW

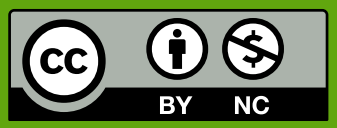


A Year in Focus: Impact Report | 2021



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A Year in Focus: Impact Report

2021

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H.E. Dr. Amna bint Abdullah Al Dahak Al Shamsi
Minister of Climate Change and Environment of the UAE



H.E. Dr. Muhammad Sulaiman Al Jasser

President of the Islamic Development Bank and Group Chairman

Acknowledgement of Core Donors

We sincerely acknowledge the Ministry of Climate Change and Environment (MOCCAE), the Islamic Development Bank (IsDB), and the Environment Agency – Abu Dhabi (EAD) for their vital and continued support. Their contributions have been integral to advancing ICBA's mission of addressing the critical challenges of climate change, water scarcity, and soil salinity in arid and saline regions. Their partnership strengthens our ability to implement impactful solutions, enhance food security, and build resilience for vulnerable communities worldwide. We deeply value their trust and collaboration, which are pivotal in achieving ICBA's vision of sustainable and climate-resilient agriculture.



H.E. Razan Khalifa Al Mubarak
Managing Director, Environment Agency - Abu Dhabi (Chair)



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Dr. Ren Wang
Director General, China National GeneBank



Dato' Dr. Mohamad Zabawi bin Abdul Ghani
Director General, Malaysian Agricultural Research and Development Institute



Dr. Tarifa Alzaabi
Director General, ICBA (Ex officio member)

Acknowledgement of the Board of Directors

We extend our deepest gratitude to the Board of Directors for their unwavering guidance, strategic vision, and commitment to ICBA's mission. Their leadership has been instrumental in advancing our work to address critical agricultural and environmental challenges in arid and saline environments. The Board's dedication to ICBA's vision of sustainable livelihoods and food security is a cornerstone of our progress and success, and we are profoundly thankful for their continued support and insightful governance.



Dr. Tarifa Alzaabi
Director General

Foreword from the Director General

The year 2021 marked a pivotal period for the International Center for Biosaline Agriculture (ICBA) as we deepened our efforts to provide resilient, sustainable agricultural solutions for communities in arid and saline environments. Building on our commitment to address global food security and environmental challenges, ICBA executed impactful, Sustainable Development Goal-aligned projects across Central Asia, the Middle East, North Africa, and sub-Saharan Africa, reaching over 3,000 beneficiaries directly.

ICBA's projects in 2021 tackled critical issues such as climate change, water scarcity, and soil salinity, working alongside rural youth in Tunisia, farmers in Egypt and the UAE, and other key stakeholders. Our initiatives have equipped communities with climate-smart agripreneurship and advanced agricultural techniques, empowering them to maximize

saline resources and adopt resilient crops like quinoa and Salicornia. These efforts reinforce our vision of achieving sustainable livelihoods and food security.

To further our scientific impact, we established the Desert Life Science Laboratory, a modern genomics research facility in partnership with BGI, the world's largest genomics institution. This facility marks a significant enhancement of ICBA's research capacity, enabling groundbreaking research in crop improvement to meet the food demands of arid regions.

Collaboration was another cornerstone of ICBA's approach in 2021, with 48 new agreements formalizing partnerships across government, academia, and industry. These alliances have amplified our reach, allowing us to leverage collective expertise to confront shared challenges and strengthen food and water security worldwide.

As we look ahead, I am deeply grateful to our partners, donors, and ICBA's dedicated team. Together, we are forging a sustainable future, innovating to provide hope and resilience for communities most affected by climate and environmental change.


ICBA

ICBA at a glance

VISION
Sustainable livelihoods and food security in marginal environments

MISSION
To work in partnership to deliver agricultural and water scarcity solutions in marginal environments

STRATEGIC OBJECTIVES

- Promote sustainable management of natural resources
- Enhance agricultural value chains
- Provide climate change solutions
- Advance sustainable food, feed & biofuel agri-technologies

ICBA is an international not-for-profit applied agricultural research center with a unique focus on marginal environments where an estimated 1.7 billion people live. It identifies, tests, and introduces resource-efficient, climate-smart crops and technologies that are best suited to different regions affected by salinity, water scarcity, and drought, among other factors.

Since its formation in 1999, the center has implemented projects in some 40 countries in Central Asia, the Middle East, North Africa, South Asia, the South Caucasus, and sub-Saharan Africa.

ICBA has also expanded its network of partners around the world to increase the reach and impact of its work. It has partners in more than 50 countries, enabling it to leverage a vast and diverse pool of expertise to achieve lasting outcomes on the ground.

It is a founding member of the Association of International Research and Development Centers for Agriculture, a nine-strong alliance focused on increasing global food security by supporting smallholder agriculture within healthy, sustainable, and climate-smart landscapes.

AIRCA
Association of International Research and Development Centers for Agriculture

Through its work, ICBA helps to create jobs, and improve livelihoods, food security and nutrition for some of the poorest rural communities around the world.

ICBA contributes to the achievement of seven Sustainable Development Goals:



2021 in numbers



74
projects





48
new collaborative agreements



267
mentions in leading news outlets



752
at 16 special technical training courses, including 277 women from 51 countries



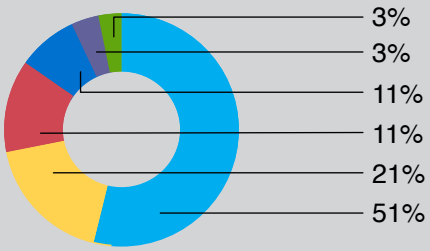
74
research publications

65 beneficiary countries*



* This also includes beneficiaries of capacity development programs.

Funding Sources



- UAE Government
- Islamic Development Bank
- Other External Donors
- International Fund for Agricultural Development
- Sergey Brin Family Foundation
- EXPO Dubai 2020



Roughly 9% of the global population lived in extreme poverty in 2021.


UNHCR



End poverty in all its forms everywhere



Young people in Tunisia are keen to start up their own companies in agriculture, but they often lack knowhow, technology, and finance.

Helping Tunisian youth venture into agribusiness

Youth unemployment is a big problem in Tunisia. A total of 38.3 percent of all young people aged 15-24 were out of work in the country in 2021*.

The problem is especially acute in parts of the country where tourism is a major earner. As a result, there are economic disparities between rural and urban areas.

While agriculture remains the main source of food and livelihood for rural communities, it does not offer young people much opportunity as it is affected by constraints like water scarcity and drought.
To make farming a profitable practice under such circumstances, they need to have appropriate skills, tools, and resources. And this is what ICBA is trying to offer under a project funded by the Albatha Group.

Titled "Creation of agricultural startups for Tunisian youth", the project is carried out by ICBA in collaboration with local authorities, women's associations, farmers' organizations, as well as private companies.

Dr. Henda Mahmoudi, a plant physiologist at ICBA and project lead, says: "The main goal of the project is, among other things, to create job opportunities for young people in Hammamet, a town in Tunisia's Nabeul Governorate. As part of the project, we are working with local partners to develop sustainable value chains for crops like quinoa and sorghum. We are also promoting almond and pistachio cultivation."

In 2021 ICBA organized several training events for young people and provided them with technical assistance and farming equipment.

The project benefitted some 85 young people, including 15 women, over its course.

* The World Bank. (2022, June). Unemployment, youth total (% of total labor force ages 15-24) (modeled ILO estimate) - Tunisia. https://data.worldbank.org/indicator/SL.UEM.1524.ZS?locations=TN

Quinoa from ICBA shows economic promise in Morocco

Agriculture is a key engine of growth in Morocco. The sector accounts for 19 percent of GDP and a third of employment*.

But it is increasingly being hampered by such factors as salinization and drought, leaving many smallholder farmers across the country in a tight spot. It is estimated that about 0.5 million hectares of the agricultural land is affected by salinization and drought is getting more frequent and severe.

To sustain their livelihoods in the face of such difficulties, more and more smallholders are turning to new technologies and crops such as quinoa. And this is already paying dividends according to a study* by a team of scientists from ICBA; the Mohammed VI Polytechnic University (UM6P), Morocco; the Hassan II Institute of Agronomy and Veterinary Sciences (IAV), Morocco; and McGill University, Canada.

The study presents different scenarios for quinoa cultivation in manual and mechanized production systems under rainfed and irrigated conditions. It notes that, among other things, quinoa is more profitable than some traditional cereals. For example, it generates much higher returns than barley and wheat when grown under the same conditions.
The study also points out that use of irrigation and organic amendment helps to increase yields by two and three times respectively compared to rainfed conditions.

What is more, mechanical pearling reduces saponin, an element which gives quinoa a bitter taste, by 68 percent as against 57 percent achieved through traditional abrasion and washing.

According to Dr. Hayatullah Ahmadzai, a social economist at ICBA and co-author of the study: *"Agricultural production in Morocco's Rehamna Province faces various challenges, including salinization and drought. The introduction of climate-resilient crops such as quinoa is a sustainable option to improve farmers' incomes, nutrition, and food security."*

The study also shares some lessons learned and recommendations based on the results of a multi-year project completed in Rehamna Province. Funded by Canada's International Development Research Centre, the project was implemented by ICBA jointly with the Mohammed VI Polytechnic University and the Ministry of Agriculture, Fisheries, Rural Development, Water and Forests of Morocco.

Under the project, five quinoa genotypes from ICBA were certified as varieties by Morocco's National Food Safety Office (L'Office National de Sécurité Sanitaire des Produits Alimentaires) and approved for commercial cultivation.

*Oumara, N. G. A. & El Youssfi, L. (2022). Salinization of Soils and Aquifers in Morocco and the Alternatives of Response. Environmental Sciences Proceedings, 16 (65). https://doi.org/10.3390/environsciproc2022016065
** Hirich, A., Rafik, S., Rahmani, M., Amira, F., Azaykou, F., Filali, K., Ahmadzai, H., Jnaoui, Y., Soulaimani, A., Moussafir, M., El Gharous, M., Karboune, S., Sbai, A. & Choukr-Allah, R. (2021). Development of quinoa value chain to improve food and nutritional security in rural communities in Rehamna, Morocco: lessons learned and perspectives. Plants, 10 (2): 301. https://doi.org/10.3390/plants10020301



Quinoa varieties provided by ICBA are doing much better in Morocco's conditions than their counterparts used by local farmers.



Farmers in Egypt and the UAE are learning how to grow food sustainably and profitably in areas affected by soil and water salinity.

Enabling farmers to earn from saline soil and water in Egypt, UAE

Many farmers in marginal areas of Egypt and the UAE have a similar dilemma. On the one hand, farming is challenging due to unfavorable environmental conditions. Yields and profits are affected. On the other hand, it is an important source of livelihood. But to produce more in these conditions, they need knowhow and technology.

To help with that, ICBA conducted a project targeting farming communities in the two countries' marginal areas. Completed in 2021, the project funded by the HSBC Bank focused on climate-smart agripreneurship and sustainable farming approaches for salt-affected regions of Egypt, and capacity development in integrated farming in the UAE.

As part of the project titled "Food for the Future II - Building Sustainable Networks and Unleashing Entrepreneurial Potential of Farming Communities Living in Marginal Areas", ICBA worked with local partners to enable farmers to make the most of salt-affected soil and water resources by cultivating quinoa and Salicornia and using integrated farming approaches.

Dr. Dionysia Angeliki Lyra, a halophyte agronomist at ICBA and project lead, says: *"One of our main objectives was to equip small-scale farmers with skills and technologies to realize their potential and become successful agripreneurs."*

For example, in Egypt, ICBA collaborated with the Desert Research Center on promoting quinoa production among farmers and developing a value chain in the New Valley Governorate, and with the Ministry of Agriculture and Land Reclamation on introducing Salicornia among farming households in the Red Sea Governorate. As a result, five quinoa and six Salicornia genotypes from ICBA were introduced on 11 farms.

Some 2,500 people were engaged in various project activities and events in Egypt. The project provided about 350 farmers, including 250 women, with different tools and resources for growing food under marginal conditions. What is more, nearly 30 quinoa- and five Salicornia-based products were launched and presented at different events.

Overall, around 3,000 stakeholders, including farmers, agripreneurs and extension staff, benefitted from capacity development, technology transfer, and other work over the course of the project.

ICBA pilots AI-powered mobile application for plant disease detection

According to some estimates, the annual loss in crop production due to pests and diseases ranges between 20 and 40 percent globally. Each year plant diseases cost the world's economy around 220 billion USD, and invasive insects around 70 billion USD*.

So it is important to develop innovative solutions like mobile applications to empower farmers and extension specialists to make prompt diagnoses. This will help to take an effective and timely action, and thus increase food security and income.

This rationale is behind a project managed by ICBA together with the University of Barcelona, Spain. Called "Developing a user-friendly application for smallholder farmers for detection of plant disorders", the project is specially designed to develop an AI-powered mobile application to help identify and deal with diseases and nutritional disorders in different crops.

Currently, the application is tailored for tomato, capsicum, cucumber, and quinoa, and its beta version is being tested in Egypt, Tunisia, and the UAE. In 2021 the application was presented to 78 farmers, extension specialists and researchers at a series of training events in three countries.

One of the participants, Eng. Mohamed Ali Al Abdouli, an agricultural extension officer at the Abu Dhabi Agriculture and Food Safety Authority, the UAE, says: "I thank ICBA for this great opportunity to attend such an important training workshop where we discussed how to use the mobile application to help farmers. Such tools are very much required to guide farmers and help them protect their crops. There is great potential to collaborate on this aspect."

Targeting farmers in Egypt, Morocco, Tunisia, and the UAE, the project aims to roll out the application in other countries as well. It is being customized for each country, and the final version will be available in three languages: Arabic, English, and French.

* FAO. (2019, April 3). New standards to curb the global spread of plant pests and diseases. https://www.fao.org/news/story/en/item/1187738/icode/



ICBA is trialing an AI-powered mobile application for detecting plant diseases and disorders among farmers and extension specialists in Egypt, Tunisia, and the UAE.







End hunger, achieve food security and improved nutrition and promote sustainable agriculture



Salicornia is a salt-loving plant that can be used for food, feed, and biofuel.

Study finds brackish water best choice for Salicornia

Fresh water is very scarce in the Middle East and North Africa. So farming is challenging. But there is plenty of saline water, with Morocco, for instance, having 2 billion cubic meters of saline groundwater*. As the region is also home to 70 percent of the world's desalination plants**, there is also a lot of reject brine.

Given that agriculture accounts for around 80 percent of water use in the region***, it is important to tap into the potential of saline water by growing halophytic, or salt-loving, plants for food and other purposes. But first, it is necessary to identify suitable crops and the most appropriate levels of water salinity for their production.

This was the purpose of a study**** by a team of scientists from ICBA and the University of Barcelona, Spain. They found that brackish water is the best option for irrigating Salicornia europaea, a multipurpose halophyte, in terms of biomass and seed production.

Published in Plant Science, the study assessed two ecotypes of Salicornia europaea that grow naturally in the coastal areas of the emirates of Ras al-Khaimah and Umm al-Quwain in the UAE.

The seeds of both ecotypes were collected in 2016 and then multiplied in ICBA's greenhouse facilities using brackish water with a salinity level of 25 deciSiemens per meter.

During the experiments, scientists used three types of water for irrigation: fresh water (0.3 deciSiemens per meter), brackish water (25 deciSiemens per meter) and sea water (40 deciSiemens per meter). They also looked into the effect of hypersaline water (80 deciSiemens per meter) on seed germination. The seeds which were given fresh water and brackish water had the highest germination rates, while use of sea water reduced the germination rate by half, and hypersaline water caused it to drop almost to zero. These findings show that brackish water produces the best outcomes for biomass and seed. Dr. Mohammed Shahid, a geneticist at ICBA and co-author of the study, says: *"Not only does the study shed light on phenotypic variability among Salicornia europaea ecotypes and provide valuable information about their agronomic and morphological traits, but it also contributes to a better understanding of this species' physiological response to salinity."*

The study proves that Salicornia europaea could be used as an alternative crop by smallholder farmers in salt-affected areas. This practice would help to reduce their vulnerability to the effects of salinization on agriculture and food security.

* Choukr-Allah, R. (2021). Use and Management of Saline Water for Irrigation in the Near East and North Africa (NENA) Region. In: K. Negacz, P. Vellinga, E. Barrett-Lennard, R. Choukr-Allah & T. Elzenga. (eds). Future of Sustainable Agriculture in Saline Environments (1st ed., pp. 51-68). Taylor & Francis Group: Boca Raton. https://doi.org/10.1201/9781003112327

** Leahy, S. & Purvis, K. (2016, September 29). Peak salt: is the desalination dream over for the Gulf states? The Guardian. https://www.theguardian.com/global-development-professionals-network/2016/sep/29/peak-salt-is-the-desalination-dream-over-for-the-gulf-states

*** UNICEF. (2021, August 22). "Running Dry": unprecedented scale and impact of water scarcity in the Middle East and North Africa. https://www.unicef.org/press-releases/running-dry-unprecedented-scale-and-impact-water-scarcity-middle-east-and-north

**** Araus, L. J., Rezzouk, Z. F., Thushar, S., Shahid, M., Elouafi, A. I., Bort, J. & Serret, D. M. (2021). Effect of irrigation salinity and ecotype on the growth, physiological indicators and seed yield and quality of Salicornia europaea. Plant Science, 304 (2021): 110819. https://doi.org/10.1016/j.plantsci.2021.110819

Promoting halophytic crops for healthy diets

In 2021 ICBA launched a new program to raise public awareness about the benefits of cooking healthy dishes using halophytic, or salt-loving, crops such as Salicornia.

Initiated under a project funded by the Expo 2020 Dubai's Expo Live Innovation Impact Grant Program and dubbed the Halophytic Kitchen Lab, the program is the first of its kind in the UAE as it aims to promote use of locally grown halophytic crops for food.

The Halophytic Kitchen Lab seeks to encourage both young people and adults to learn more about the health and other benefits of consuming foods based on nutritious halophytic crops which are cultivated with saline water in desert conditions. Participants are introduced to the art of cooking and devising healthy recipes of dishes with halophytic crops under the guidance of a professional chef.

The program also includes such activities as lectures on challenges to food security and the importance of halophyte-based food products, as well as tours of ICBA's research stations.

According to Dr. Dionysia Angeliki Lyra, a halophyte agronomist at ICBA: "Halophytes have great potential as alternative crops for food production in semi-arid and arid regions such as the Middle East and North Africa where farming faces constraints ranging from soil and water salinity to water scarcity."

But several bottlenecks should be eliminated before such crops become widely cultivated and used. They include, among others, weak value chains and low or non-existent consumer awareness and demand.

For this reason, ICBA has been focusing more and more efforts on developing value chains for Salicornia and quinoa in different countries, including Egypt, Morocco, and the UAE.

With the start of the Halophytic Kitchen Lab, ICBA will facilitate the development of value chains for halophytic crops, informing consumers and thus stimulating demand for halophytic products.

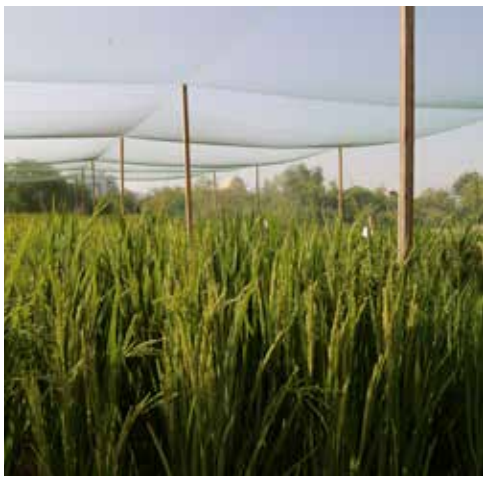


Salicornia is a nutritious halophyte that can be used as a vegetable in salads and other dishes.















Soil and water salinity is a major constraint on crop growth.

Scientists look for ways to grow salt-tolerant rice in UAE

Staple crops like rice are usually not up to the job when it comes to salt tolerance. Yet there are some exceptions as certain varieties can withstand moderate levels of salinity if conditions are right.

This is what scientists at ICBA tried to find out as part of an experiment completed in 2021. In particular they searched for the most suitable irrigation method to grow salt-tolerant rice under saline and desert conditions in the UAE. They studied efficient approaches to cultivating salt-tolerant rice in the local environment.

In the trial at ICBA's research station between September 2020 and May 2021, scientists evaluated four salt-tolerant rice varieties from the Bangladesh Rice Research Institute (BRRI): BRRI dhan 47; BRRI dhan 61; BRRI dhan 67; and BRRI dhan 78.

The varieties were tested using three irrigation methods: sub-surface drip irrigation without plastic lining; sub-surface drip irrigation with a sealed iron sheet at the bottom (a weighing-type lysimeter); and surface drip irrigation with plastic lining.

Results show that a grain yield of about 7.5 tonnes per hectare was achieved when surface drip irrigation with plastic lining was used. The medium- and long-duration varieties flowered slightly late due to cold spells during December and January and 50-percent flowering was recorded within 118-154 days. What is more, BRRI dhan 78, a late-flowering variety, performed relatively better when surface drip irrigation with plastic lining was used compared to the other varieties, followed by BRRI dhan 61, a medium-duration variety.

Dr. RK Singh, Program Leader on Crop Diversification and Genetics at ICBA, says: "Our results showed that short-duration varieties can perform relatively better under local conditions. I think rice cultivation could be scaled up in the country in the future after the most appropriate practices for the local environment have been identified and standardized."

Overall, the findings suggest that rice's response to marginal conditions varies according to sowing dates. For example, a higher yield can be obtained if seeds are sown in the second week of September in a controlled environment facility (a greenhouse or net-house), followed by transplanting seedlings to the open field in the first week of October.

As a follow-up to the experiment, ICBA began a large-scale study using water at different salinity levels to identify varieties suitable for the UAE and other countries with similar environments and estimate the exact water requirements for varieties under different stresses.

New genomics facility opens up opportunities for advanced breeding research

In 2021 ICBA became home to a new advanced genomics facility. It is designed to further breeding research on, among other things, healthier, more nutritious, and resilient crops.

Called the Desert Life Science Laboratory, it is an outcome of the strategic partnership between ICBA and BGI, the world's largest genomic research institution.

The facility will help to fast-track the discovery and development of food and other crops suited to marginal environments. It will also serve the needs of both public and private entities for various types of sequencing services in the UAE and other countries.

Its opening marks an important milestone in ICBA's efforts to address agricultural challenges through use of cutting-edge science and technology.

H.E. Mariam bint Mohammed Almheiri, Minister of Climate Change and Environment of the UAE, says: *"The Desert Life Science Laboratory is a vital initiative and one that is a significant addition to the UAE's food security efforts, as enshrined in the country's National Food Security Strategy 2051. Increased desertification as a result of climate change is expanding marginal environments, so finding crops with a food value that can not only tolerate but thrive in these environments is a strategic priority."*

The laboratory is intended to carry out the complete gamut of genomic research from genome-wide association studies to soil metagenomic analysis. It is specially built to conduct whole genome resequencing, genotyping-by-sequencing, metagenomics, transcriptomics, and other omics studies. It is equipped with some of the latest biotechnological equipment, including the next-generation sequencing platform (DNBSEQ-G400RS); M220 Focused-ultrasonicator™; Agilent Bioanalyzer 2100; and the Applied Biosystems™ Real-time PCR. The laboratory also features such tools as a Qubit 4 Fluorometer; PCR machines (Bioer and Applied Biosystems™); high-speed centrifuges; an autoclave; laminar flow hoods; and growth chambers.

Not only will the laboratory facilitate breeding research, but it will also act as a hub for knowledge transfer. Researchers, specialists, and students from the UAE and abroad will have opportunities to improve their knowledge and gain practical skills by participating in various genomic experiments. What is more, the laboratory will be able to help to protect local plant breeders' rights by providing services in plant variety registration, identification, and validation.

Ultimately, it will bolster ICBA's research and development capacity for developing plants which produce more food with fewer resources to boost food security and nutrition in marginal environments.



The Desert Life Science Laboratory was inaugurated by H.E. Mariam bint Mohammed Almheiri, Minister of Climate Change and Environment of the UAE.



Women form the backbone of the rural economy in regions like sub-Saharan Africa and North Africa.





Achieve gender equality and empower all women and girls

Regional program for Arab women scientists welcomes new fellows

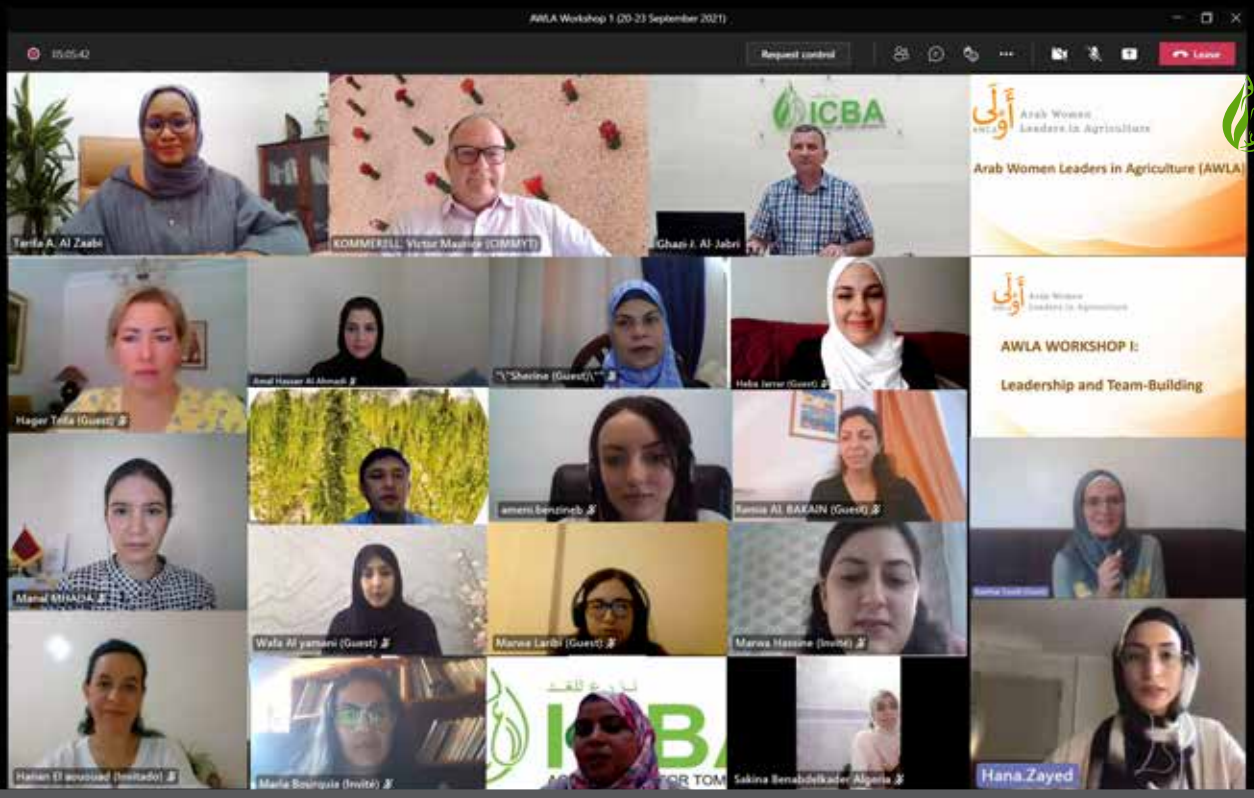
In 2021 ICBA started the second edition of the Arab Women Leaders in Agriculture fellowship program to continue building up its impact and reach.

After a competitive selection process, a group of 19 women researchers from seven countries joined the ten-month-long program. The second cohort included women professionals from different backgrounds and countries such as Algeria, Egypt, Jordan, Lebanon, Morocco, Tunisia, and the UAE.

Managed by ICBA, the second edition was funded by the Bill & Melinda Gates Foundation and the CGIAR Research Programs on Maize and Wheat.

Dr. Tarifa Alzaabi, Director General of ICBA, says: "Women-led contributions to agriculture, both on the farm and in the lab, are essential components of global food security. In the Middle East and North Africa, women make up over 40 percent of the labor force in the sector. They also contribute around 18 percent to the region's overall GDP. Through our program, we hope to encourage women to play a more active role in such areas as science and innovation for sustainable agricultural development and food security."

Being the first of its kind, the program facilitates Arab women researchers' access to leadership roles, provides platforms to showcase their intellect, capability, and contribution, and encourages gender-responsive working cultures and enabling environments. It also contributes to the achievement of several Sustainable Development Goals. Its long-term goal goes beyond capacity development and includes improved food security and nutrition, a better research and development landscape, and economic and social benefits of a narrowed gender gap in the region.



The Arab Women Leaders in Agriculture fellowship program is focused on narrowing the gender gap in agricultural science and leadership in the Middle East and North Africa region.



Afef Ladhari
Tunisia



Amal Hassan Al Ahmadi
UAE



Ameena Ali Kulaib Al Tenaiji
UAE



Ameni Ben Zineb
Tunisia



Hager Snoussi
Tunisia



Hana Zayed
Egypt



Hanan El Aou-Ouad
Morocco



Heba Jarrar
Jordan



Kawthar Esseili
Lebanon



Khalwla Mohammed Darwish Al Marzooqi
UAE



Manal Mhada
Morocco



Maria Bourquia
Morocco



Marwa Laribi
Tunisia



Mouza Saeed Mohammed Al Shehhi
UAE



Noura Saeed Mohammed Al Nuaimi
UAE



Ramia Al Bakain
Jordan



Rana Jreij
Lebanon



Sakina Benabdelkader
Algeria



Wafa Al-Yamani
UAE



ICBA places special emphasis on giving women skills and resources to increase their contribution to agriculture and food security.

Empowering women agripreneurs in UAE

As ICBA puts women at the center of its capacity development efforts, it joined forces with the Ajman Business Women Council, an affiliate of the Ajman Chamber of Commerce and Industry, the UAE, in 2021 to provide knowledge and consultation to local women to become successful agripreneurs.

Under this collaboration, ICBA will increase the range and reach of its capacity development offerings and help to boost agripreneurship not only in the UAE but also in other countries. This work is also aligned with the center's programs like the Arab Women Leaders in Agriculture.

In particular the two organizations will work together to encourage businesswomen to pursue agricultural projects through the Plant and Reap, one of the council's main initiatives for 2021. This will help to generate more interest among women in the agricultural sector and eventually support food security at the national level.

Dr. Amna Khalifa Al Ali, Chairwoman of the Ajman Business Women Council, says: "AJBWC is keen to keep pace with the mechanisms required to support the agricultural sector and food security by guiding women entrepreneurs to enter this field, especially given the significant agricultural progress the country is witnessing using modern agricultural models and promising innovative tools for successful agricultural projects."

ICBA and the Ajman Business Women Council will also develop projects and capacity development programs, including training workshops focused on agripreneurship targeting, among others, council members.

They will also offer agricultural consultations under the Plant and Reap initiative which is aimed at encouraging businesswomen to achieve self-sufficiency in agricultural products.







Ensure availability and sustainable management of water and sanitation for all



Irrigated agriculture is a major contributor to GDP in Central Asia.

Shedding new light on water-energy nexus in lift irrigation schemes in Central Asia

Agriculture is irrigated across much of Central Asia. And different methods of irrigation are used to cultivate crops. But each method has its own pros and cons, with some having far-reaching consequences.

In lift irrigation, for instance, external power, including fuel-based or electricity-powered pumps, is necessary to take water to the fields. As a result, there is competition for water and energy for different purposes.

A study* by a team of scientists from ICBA and several institutions in Uzbekistan looked into what impact lift irrigation has on water and energy use in the region. Published in Renewable and Sustainable Energy Reviews, it presents some insights with regional implications. According to Dr. Aziz Karimov, Head of ICBA's Regional Office for Central Asia and the South Caucasus and co-author of the study: "Overall, we focused on two key objectives in this study: evaluate transboundary water-energy nexus effects on emerging water and energy issues; and examine aquifer storage and recovery and drip irrigation in lift irrigation areas for more productive use of water and energy resources."

Lift irrigation schemes occupy more than 50 percent of the irrigated land in Uzbekistan and 40 percent in Tajikistan across the Aral Sea basin. But they require external power to transport water from different sources to reservoirs for agriculture and other needs.

In this context, a shift from hydropower to thermal power supply for lift irrigation has direct and indirect negative consequences. This, in turn, leads to a growing competition for water use between agriculture and energy generation.

Moreover, problems along the agriculture-energy nexus are on the rise because of inadequate cooperation among countries over transboundary river basins. For example, separating downstream countries from upstream hydropower generation is resulting in the development of alternative water and energy sources to accommodate the high demand for water resources.

While causing water scarcity, this approach also accelerates the depletion of fossil energy resources and increases greenhouse gas emissions, thus complicating the overall management of water resources at the national level. The study suggests that under these circumstances, it is important to increase water and energy efficiency and productivity, including the adoption of renewable energy, and to shift from conventional furrow irrigation to groundwater-based drip irrigation.

It concludes that despite the potential for significant improvement of water and energy productivity at the national level, there is no sustainable alternative to meeting the increasing demands of the population and the environment other than benefit-sharing in water and energy and enhancing cooperation across river basins in Central Asia.

* Karimov, K. A., Toshev, R. H., Karshiev, R., Karimov, A. A. (2021). Water–energy nexus in Central Asia's lift irrigation schemes: Multi-level linkages. Renewable and Sustainable Energy Reviews, 147 (2021): 111148. https://doi.org/10.1016/j.rser.2021.111148

Supporting sustainable water management in Bahrain

Water scarcity is a big problem in Bahrain. In fact it is ranked 12th on the list of the most water-stressed countries in the world*.

So sustainable water management is a top priority for the national government. And the country is focusing on improving institutional capacities in integrated water resources management.

In 2021 ICBA signed a cooperative agreement with the National Oil and Gas Authority to support the country's efforts in this field.

Under this collaboration, ICBA will provide technical assistance for ongoing activities to strengthen the Water Resources Council, a Cabinet-level inter-ministerial organ of the government responsible for coordinating water policy formulation and implementation. This will help to broaden the council's functions and enhance its role in climate risk management and wastewater recycling, treatment, and reuse.

H.E. Mr. Nasser Sultan Al-Suwaidi, CEO of the National Oil and Gas Authority (now part of Bahrain's Ministry of Oil), says: *"This agreement was signed in the context of the Kingdom's water sector development, with the project's scope including the development of institutional capacities of government sector employees in integrated water resources management (IWRM), which will aid in the management of water resources as a whole, which is considered one integrated network."*

The collaboration is also designed to develop government departments' capacities to implement climate change adaptation strategies across the water sector.

The overarching goal is to equip relevant institutions with the means to conduct water resources monitoring and assessment; generate information and knowledge using relevant analytical tools and models; and provide evidence to support decision-making processes. This work is part of the project named "Development of an Enabling Environment for Water Demand Management", which is financed by the Green Climate Fund and is managed by the government of Bahrain through the National Oil and Gas Authority.

* World Resources Institute. (2019, August 6). 17 Countries, Home to One-Quarter of the World's Population, Face Extremely High Water Stress. https://www.wri.org/insights/17-countries-home-one-quarter-worlds-population-face-extremely-high-water-stress





Improving water accountability, water use efficiency and water conservation are prerequisites for achieving water security in Bahrain.



Treated sewage effluent can serve as an alternative to scarce fresh water in the UAE.

Harnessing treated sewage effluent's potential for food and water security in UAE

The UAE is considered one of the most water-stressed countries globally*. Yet it has one of the world's highest per capita water consumption rates. On top of that, agriculture gets the lion's share of fresh water - about 72 percent - compared to the municipal and other sectors. But despite its large water footprint, the agricultural sector supplies under 20 percent of the nation's food**.

So the nation has been looking for other sources for irrigation like treated sewage effluent. And this idea is behind a new project commissioned by the Ministry of Climate Change and Environment of the UAE.

Led by ICBA and implemented jointly with the International Center for Agricultural Research in the Dry Areas; the United Arab Emirates University; and the Food and Agriculture Organization of the United Nations, the project aims to explore how controlled environment agriculture systems such as greenhouses and vertical farms can benefit from using treated sewage effluent and biosolids. Controlled environment agriculture is one way to cut down on water use as it makes it possible to grow crops with up to 95 percent less water than traditional agricultural methods. It is also one of the main planks of the UAE's National Food Security Strategy 2051.

H.E. Mariam bint Mohammed Almheiri, Minister of Climate Change and Environment of the UAE, says: "The UAE's annual water demand amounts to around 5 billion cubic meters, with treated sewage effluent accounting for about 11 percent of this demand. We produce around 735 million cubic meters of treated sewage effluent each year, but more than a quarter of this resource is either lost or discharged into the sea. This is a tremendous waste, as treated sewage effluent is rich in nutrients, making it an ideal medium to support crop growth."

Reuse of treated sewage effluent will help to reduce pressure on freshwater resources and ensure sustainable water management. And as large amounts of biosolids are produced every year, they will help to replace artificial fertilizers.

More importantly, they will add value to farming as more food will be produced without freshwater resources or artificial fertilizers.

It is hoped that the project will pave the way for safe and effective use of treated sewage effluent and biosolids for agriculture in the country, thus contributing to national food and water security efforts.

* World Resources Institute. (2019, August 6). 17 Countries, Home to One-Quarter of the World's Population, Face Extremely High Water Stress. https://www.wri.org/insights/17-countries-home-one-quarter-worlds-population-face-extremely-high-water-stress

** Solomon, E. (2016, February 23). How Agricultural Waste Can Contribute to the UAE's Food, Water and Energy Security. Khalifa University. https://www.ku.ac.ae/how-agricultural-waste-can-contribute-to-the-uaes-food-water-and-energy-security#:~:text=The%20UAE%27s%20agricultural%20sector%20is%20a%20significant%20user%20of%20the,20%25%20of%20the%20UAE%27s%20food

Solar-powered small-scale irrigation offers farmers hope in sub-Saharan Africa

Irrigation is a big hurdle for the majority of farmers in sub-Saharan Africa who usually own small patches of land. What they produce is enough to feed their families and sometimes sell. But because precipitation is low and agricultural conditions are poor, irrigation means everything.

A multi-year study* by scientists from ICBA offers some hope to these farmers. It shows that solar-powered small-scale irrigation systems can be an effective and low-cost solution to improve agricultural productivity and income.

Based on the results of a project titled "Scaling up small-scale irrigation technologies to improve food security in sub-Saharan Africa", the study notes that these systems can help farmers to water their lands and boost crop yields and profits.

Financed by the OPEC Fund for International Development, the project led by ICBA was implemented from 2016 to 2020 in four countries: Burkina Faso, Niger, Mali, and Senegal.
Under the project, scientists evaluated drip, sprinkler, and Californian irrigation systems to determine their technical and economic efficiency. These systems were operated using solar-powered pumps to take water from streams and wells to farmers' fields. Several major vegetables were grown across all locations and the performance of the systems was compared with that of the traditional bucket and gravity irrigation methods.

The study found that the Californian and drip irrigation systems fared much better in terms of water use efficiency and yield as opposed to all conventional irrigation methods smallholder farmers use in these countries. What is more, the economic efficiency and water productivity of these systems were two to three times higher than those of the traditional irrigation systems.

The study also found that most of the smallholder farmers preferred the solar-powered Californian irrigation system because of its relatively low installation cost (600-1,000 USD per hectare) compared to the drip irrigation system that requires about 2,000-4,000 USD per hectare as an initial investment.

The study revealed a significant impact on water consumption. For example, the total amount of water for tomatoes irrigated using the Californian irrigation system was 18 percent lower than that of the gravity irrigation method. The findings also indicated that smallholder farmers could earn up to three times more profit for vegetables grown with the solar-powered irrigation systems.

According to Dr. Asad Qureshi, a senior scientist in irrigation and water management at ICBA and project lead: *"Scaling up small-scale irrigation technologies with affordable access to energy should be a top priority in sub-Saharan Africa where despite ample water resources, only about 2 percent of the total renewable water resources are used for irrigation due to a lack of water infrastructure."*

In conclusion, the study highlighted some key policy lessons for irrigation development in the region. In particular it emphasized the need for governments to recognize the importance of irrigation development for enhancing food security and economic growth. The governments also need to extend credit facilities and soft loans to smallholder farmers to invest in these technologies.

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* Qureshi, S. A., Traore, A., Kane, M. A., Zakari, H. A. & Cisse, M. (2021). SCALING UP SMALL-SCALE IRRIGATION TECHNOLOGIES FOR IMPROVING FOOD SECURITY IN SUB-SAHARAN AFRICA (FINAL REPORT). International Center for Biosaline Agriculture. https://www.biosaline.org/sites/default/files/Project-Report-Final-OFID-October-2021.pdf











Many farmers in sub-Saharan Africa cultivate small plots of land and depend on irrigation.



2021 saw new records for four key climate change indicators.





Take urgent action to combat climate change and its impacts



Climate-resilient crops like quinoa hold great potential for food security and nutrition in marginal environments.

Turning spotlight on world's indigenous peoples in climate action

Indigenous peoples around the world – the guardians of traditional knowledge systems and underutilized, forgotten crops that may yet prove critical for global food security – are likely to be hit hardest by climate change, according to a United Nations Food Systems Summit 2021 brief* co-authored by scientists from ICBA.

More worryingly, the invaluable knowledge they have passed on from generation to generation may be lost, along with the many indigenous plants they have been the stewards of for thousands of years.

Inhabiting regions as diverse as arid and semi-arid plains, rainforests, and mountains – collectively referred to as marginal environments – indigenous peoples have a wealth of traditional practices that should form an integral part of efforts to make the global food system more resilient to climate change and other threats.

However, these assets are currently at risk of being lost owing to a range of factors.

Dr. Hayatullah Ahmadzai, a social economist at ICBA and co-author of the study, says: "Indigenous food systems and the knowledge associated with such systems are under threat today. Nearly 10 percent of all domesticated breeds of animals for food and agriculture are already extinct and another one million plant and animal species now face extinction."

While this agrobiodiversity is the bedrock of more resilient and sustainable food systems, it is often sidelined by major crops, which receive disproportionately more investment, research and development support and policy attention.

Of over 30,000 edible plants, fewer than 30 provide today most of the food consumed by 7.8 billion people.

Mainstream crops do well on arable land, but they fare badly in marginal environments. Nonetheless, mainstream agricultural policies favor staple crops over their indigenous counterparts even in marginal areas.

As a result of strategies promoting monocropping and market-oriented production, indigenous food systems are being overtaken by major cash crops. Yet the gap between the actual and potential yields of mainstream crops is widening and agricultural production continues to decline.

More troublingly, hunger and poverty in indigenous communities has been on the rise.

Evidence shows that targeted policy support and investment for indigenous food systems can turn around this situation by raising productivity, improving nutrition, and reducing carbon emissions.

To achieve food security and sustainable livelihoods in marginal areas, it is imperative to focus actions and investments on several priorities, including evidence-based policies and coordinated investments.

All this will help to transform indigenous food systems and ensure food security and sustainable livelihoods in marginal areas.

The paper concludes that while scientists and policymakers may be tempted to search for answers to hunger and poverty in technological breakthroughs, indigenous peoples and their knowledge are likely to hold the key to global challenges like climate change.

* Azam-Ali, S., Ahmadzai, H., Choudhury, D., Goh, V. E., Jahanshiri, E., Mabhaudhi, T., Meschinelli, A., Modi, T. A., Nhamo, N. & Olutayo, A. (2021, April 5). Marginal areas and indigenous people: Priorities for research and action. The United Nations Food Systems Summit 2021. https://sc-fss2021.org/wp-content/uploads/2021/04/FSS_Brief_Marginal_areas_indigenous_people.pdf







Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

Flagging critical knowledge gaps in global soil salinity research

There are a number of critical knowledge gaps and priorities in global soil salinity research according to a study* by a team of scientists from ICBA and other institutions from several countries.

Published in Advances in Agronomy, a book series, the study led by the University of California, Davis, reviews the status of global soil salinity research and analyzes the progress in combating soil salinization in various parts of the world.

In particular the study singles out ten priorities in global soil salinity research such as, among others, the need for soil salinity mapping; the need for a broader understanding of physiological mechanisms for adaptation to saline soils; the application of remote sensing to detect and map soil salinity; the need for better knowledge of the combined and interactive effects of drought and salinity stress; and the reassessment of crop salt tolerance.

The authors offer recommendations for tackling the impact of soil salinization on agriculture. For example, they note that it is important to cross advanced breeding lines in field conditions for a range of salinity and water deficit levels to find more salt-tolerant crop species.

Dr. Asad Qureshi, a senior scientist in irrigation and water management at ICBA and co-author of the study, says: "It is important to accelerate efforts to breed improved crop varieties with resistance to salinity as soil salinization has been on the rise. Around 1 billion hectares of land is estimated to be affected by salinity and sodicity worldwide to varying degrees."

It presents case studies from different countries, including Australia, China, and Pakistan, as well as specific examples from the Indo-Gangetic basin, Nile basin, Euphrates and Tigris basin, among others.

It also provides detailed insights into the extent of soil salinity around the world and gives an outlook for its impact on food security.

The authors warn that the reduced availability of freshwater resources, the continued decrease in agricultural land, climate change, and other factors could exacerbate food insecurity in the world's salt-affected areas.

Overall, the study adds to the growing body of research on global soil salinity and aims to draw attention to the risks of failure to mitigate its impact. It also urges the allocation of more funding to develop innovative solutions to address the impact of soil salinity.

* Hopmans, W. J., Qureshi, S. A., Kisekka, I., Munns, R., Grattan, R. S., Rengasamy, P., Ben-Gal, A., Assouline, S., Javaux, M., Minhas, S. P., Raats, C. P. A., Skaggs, H. T., Wang, G., De Jong van Lier, Q., Jiao, H., Lavado, S. R., Lazarovitch, N., Li, B. & Taleisnik, E. (2021). Chapter One - Critical knowledge gaps and research priorities in global soil salinity. In: D. L. Sparks. (eds). Advances in Agronomy (169: 2021, pp. 1-191). Academic Press. https://doi.org/10.1016/bs.agron.2021.03.001



Roughly 30 percent of the world's irrigated areas are salinized.



Marginal lands can play a key role in enhancing rural livelihoods around the world.

Defining policy priorities for world's marginal lands

Marginal lands – areas of the world with serious constraints on agriculture – should receive more research and policy attention in global efforts to improve food security and reduce poverty according to a study* by a team of scientists from ICBA and the Food and Agriculture Organization of the United Nations.

Published in Sustainability, a journal, the study reviews the status of the world's marginal lands and their role in food security, poverty reduction, and environmental sustainability.

More importantly, it identifies research and policy priorities for marginal lands with a focus on poverty reduction, preservation of biodiversity, and restoration of marginal lands, using strategies focused on the food-poverty-environment nexus.

What is more, the study presents a roadmap for better research engagement and provides an outlook and policy options to optimize the use of these lands.

To identify the extent of global and regional marginal lands, the study presents a working definition for agricultural lands that are referred to as marginal in the context of a given agricultural economy.

Dr. Hayatullah Ahmadzai, a social economist at ICBA and co-author of the study, says: "It is important to understand agricultural marginality in general and identify agricultural areas that are considered marginal for future research and policy interventions."

The study estimates that marginal areas make up 15 percent of the current agricultural lands worldwide and 21 percent of the total global land resources. Nearly one-third of the world's rural population live in these areas, with most of them being the poor from developing nations.

The study also stresses that agriculture in marginal areas is the primary source of food and livelihood for the majority of the world's poor and will play an increasingly important role in feeding the future population. But issues like erosion, salinization, and waterlogging are leading to declining yields.

The authors point out that marginal lands are essential for producing sufficient food to feed growing populations and keep food prices affordable. They note that investments in these areas could be a win-win solution for boosting agricultural productivity, as well as addressing global poverty.

They conclude that policies for marginal areas should integrate data generation and knowledge sharing and suggest that land tenure reform may be needed to ensure more secure and equitable land access and property rights for farmers living in marginal areas.

* Ahmadzai, H., Tutundjian, S. & Elouafi, I. (2021). Policies for Sustainable Agriculture and Livelihood in Marginal Lands: A Review. Sustainability, 13 (8692). https://doi.org/10.3390/su13168692

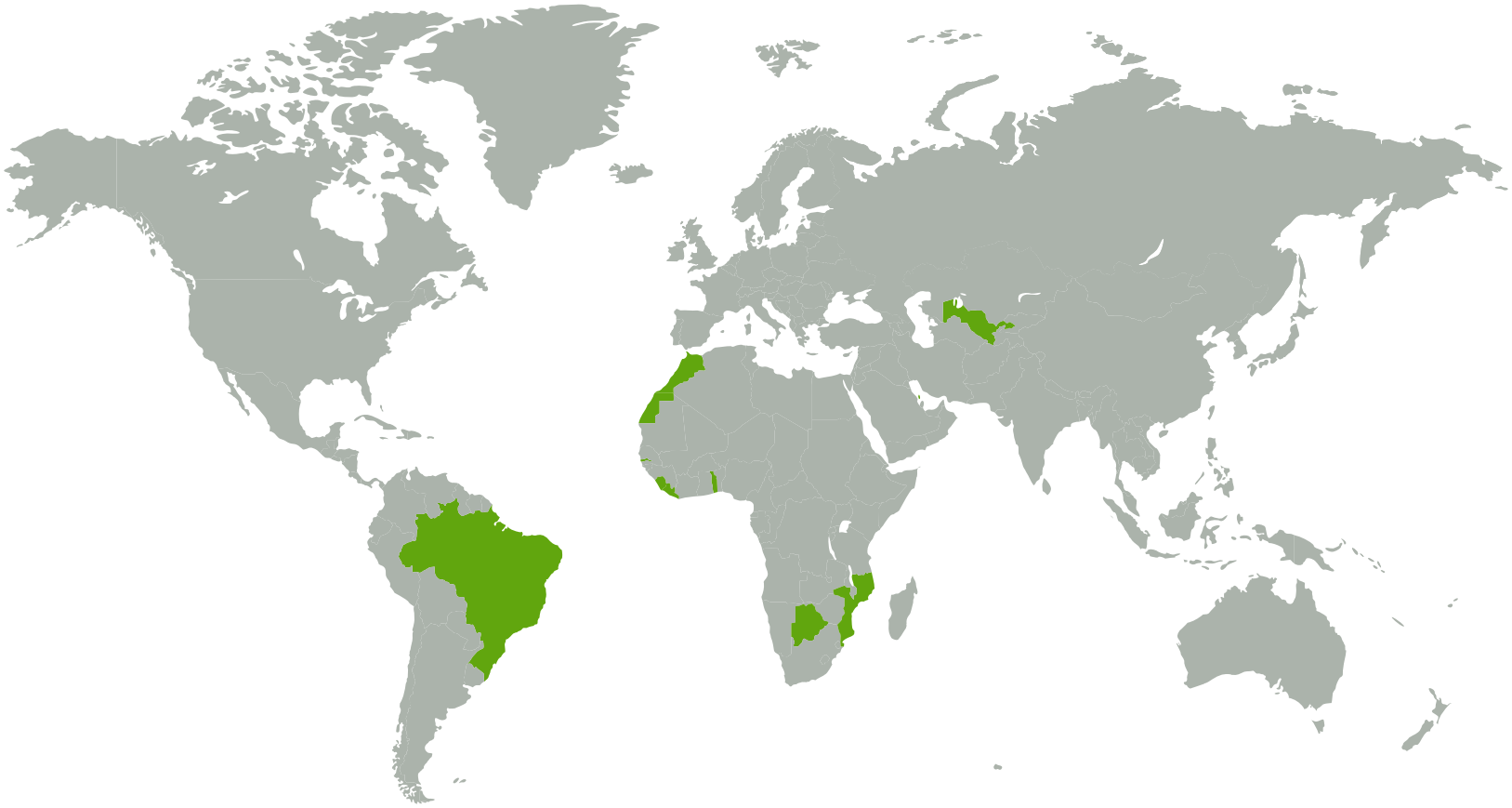


Sharing plant genetic resources

Under its mandate for marginal environments, ICBA collects, preserves, and shares germplasm of plant species with proven or potential salinity, heat, and drought tolerance.

ICBA provides seed samples to different institutions around the world for research, breeding, and introduction.

In 2021 the center provided 337 seed samples belonging to 14 different crops to partner organizations and farmers in ten countries.



Partnerships of every kind must be at the heart of the Sustainable Development Goals.







Strengthen the means of implementation and revitalize the global partnership for sustainable development

Partnering for institutional capacity development in Southeast Asia

In 2021 ICBA signed a memorandum of understanding with the Southeast Asian Regional Center for Graduate Study and Research in Agriculture to collaborate on, among other things, capacity development and knowledge sharing in Southeast Asia.

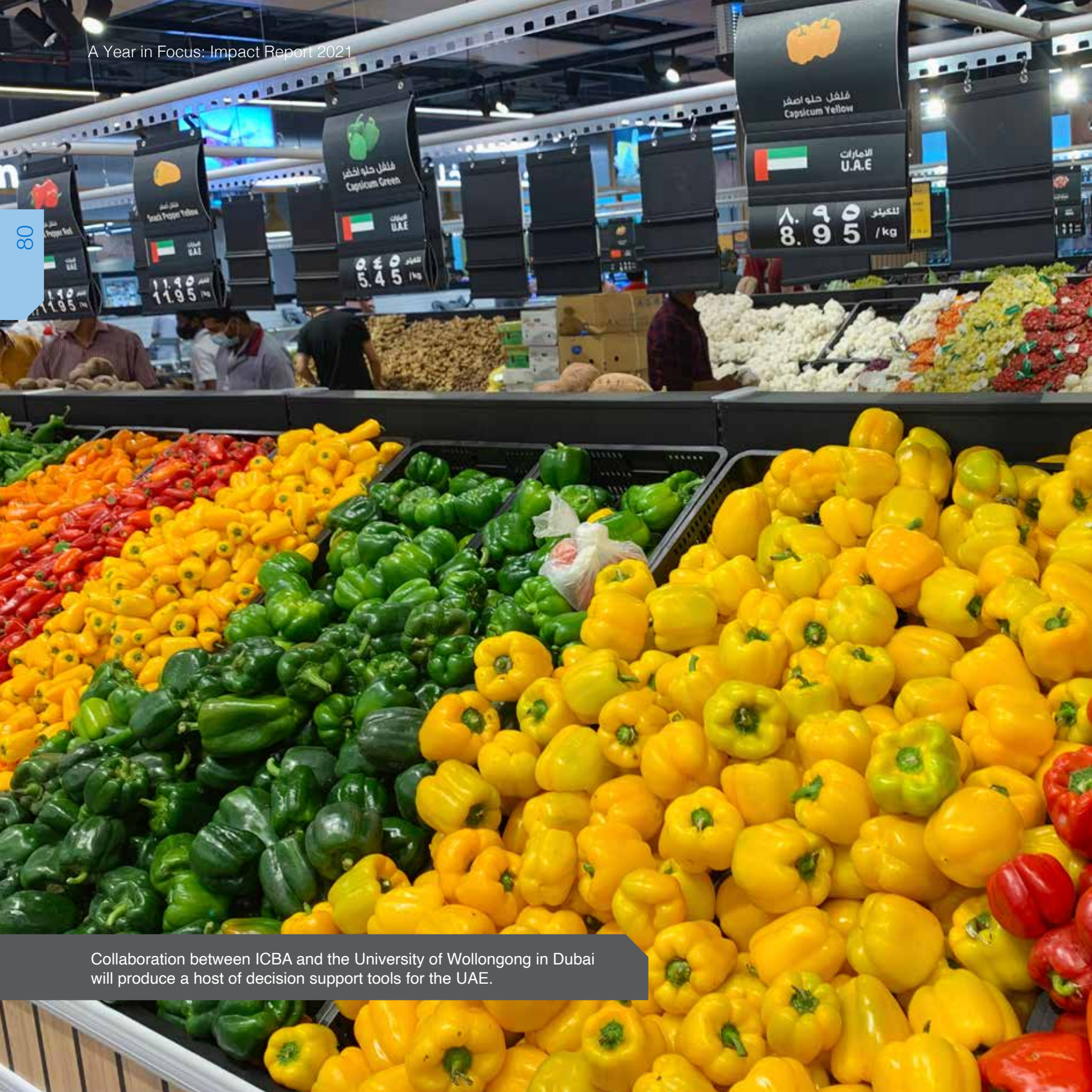
Under the agreement, the two organizations aim to develop individual and institutional capacities, as well as exchange scientific materials, publications, and information.

Dr. Tarifa Alzaabi, Director General of ICBA, says: "Partnership is a keyword in ICBA's mission statement. That is why we are really pleased to form this partnership with SEARCA. By tapping into each other's strengths, our organizations will be able to create new capacity development opportunities for researchers, especially youth and women. We also look forward to extending the reach of our capacity development programs to Southeast Asian countries."

Through this partnership, ICBA will also share its technical knowhow in Southeast Asian countries in such areas as management of natural resources; climate change modeling and adaptation; crop improvement and sustainable production; and integrated agri-aquaculture systems.



The partnership between ICBA and the Southeast Asian Regional Center for Graduate Study and Research in Agriculture makes it possible to extend the reach and scope of ICBA's capacity development offerings to Southeast Asia.



Collaboration between ICBA and the University of Wollongong in Dubai will produce a host of decision support tools for the UAE.

Joining forces to support food system resilience in UAE

In 2021 ICBA and the University of Wollongong in Dubai agreed to combine their expertise to develop a suite of innovative tools for bolstering the resilience of the UAE's food system.

Under a project titled "Resilient agrifood dynamism through evidence-based policies", the two institutions will cooperate on providing policymakers with a holistic analytical framework for monitoring and assessing a wide range of factors impacting food security at the national, regional, and global levels.

Harnessing the power of big data, the project will produce, among other things, a platform for evidence-based policymaking and a system for assessing vulnerabilities and responding to risks to food security.

ICBA and the University of Wollongong in Dubai will also formulate policies and actions to help make the country's food system more resilient to supply shocks and disruptions. What is more, the project will design a dedicated knowledge portal and a database, as well as create a set of scenarios and models.

Professor Mohamed Salem, President of the University of Wollongong in Dubai, says: "As an innovative and forward-thinking higher education provider in the UAE, UOWD is proud to partner with ICBA on the READY project, a future-focused initiative that aims to strengthen food security in the country. As an arid country that traditionally relied on imports to meet its food requirements, food security and diversifying food sources are crucial priorities for the UAE with the country seeking to address its myriad challenges in this space at scale. Through leveraging the resources of our research department, UOWD is pleased to facilitate data and analyses to enable decision making, policy development, and strategic planning processes. In doing so, our collaboration with ICBA will empower the concerned government departments to enhance local agricultural practices and ensure the sector's robustness under all circumstances."

The project will also develop an early warning system which will help policymakers anticipate potential challenges and take proactive measures. More importantly, it will offer a roadmap for transitioning the UAE's food system to a more energy-efficient, sustainable, and resilient model. Ultimately, it will contribute to broader efforts on food security and self-sufficiency in the country and benefit society as a whole.

Boosting food security efforts in Organisation of Islamic Cooperation member countries

In 2021 ICBA formalized its collaboration with the Islamic Organization for Food Security, a specialized institution of the Organisation of Islamic Cooperation, to support efforts on sustainable livelihoods and food security in Organisation of Islamic Cooperation member countries. The two organizations signed a memorandum of understanding to this effect and agreed to work together on water and land management technologies; remote sensing and modeling for climate change adaptation; crop productivity and diversification; aquaculture; and bioenergy.

Mr. Yerlan A. Baidaulet, Director General of IOFS, says: "I am of the robust opinion that our bilateral partnership will create a great synergetic effect in the mission of achieving the zero-hunger goal."

The agreement also envisages knowledge exchange, including the sharing of research studies on a range of topics, to improve agricultural productivity and sustainability.

What is more, the two organizations will focus on promoting a value-chain approach for agricultural development to facilitate agribusiness activity and market access. They will also work on developing agribusiness incubators and stimulating the creation of competitive agribusinesses.



ICBA will work with the Islamic Organization for Food Security to share its technical knowhow and scientific expertise.



ICBA and Astrum IT Academy aim to provide young people in Uzbekistan with skills and opportunities in digital agriculture.

Creating new opportunities for youth in Uzbekistan

In 2021 ICBA formed a partnership with Astrum IT Academy, an international IT academy based in Uzbekistan, to cooperate in, among other things, capacity development and knowledge sharing targeting specifically young people.

The partnership focuses on joint work in such areas as data-driven agriculture and agri-digitization. It also envisages the provision of opportunities to the academy's students to engage in ICBA-led initiatives like the ICBA Youth Engagement Society which is designed to connect young people in different countries.

Under the partnership, the two organizations will also offer internships to students, develop proposals for capacity development programs, and organize joint conferences and workshops. What is more, they will work on localizing software solutions into Uzbek and other Central Asian languages.

Dr. Tarifa Alzaabi, Director General of ICBA, says: "At ICBA we pay special attention to empowering youth in different countries through our tailor-made programs like the ICBA Youth Engagement Society. So we are delighted to join forces with Astrum IT Academy to offer capacity development opportunities to young people in Uzbekistan and extend the reach of our youth-focused programs to Central Asia."

Through this partnership, ICBA intends to share its technical knowhow with young people in Uzbekistan and other Central Asian countries with an overall goal to support efforts to address local and global challenges related to food security and agricultural production and contribute to the achievement of the Sustainable Development Goals.




STATION -1
DATE PALM

Knowledge sharing

One of the main outputs of ICBA's work is knowledge. The center is committed to generating and sharing this knowledge with all stakeholders from smallholder farmers to policymakers. ICBA disseminates science-based knowledge by means of capacity development, knowledge hubs, and communications.



Young people are at the heart of ICBA's capacity development programs.

Raising youth awareness about sustainable agriculture in UAE

With its continued focus on equipping young people with skills and resources to become champions of sustainable development, ICBA hosted its first winter camp in 2021 for schoolchildren aged between 10 and 16 to promote agripreneurship and raise awareness about sustainable agriculture in the UAE.

Organized jointly with the Dubai Entrepreneurship Academy and the Gracia Group, the winter camp gave the students an opportunity to get a better understanding of farming and agribusiness, as well as various tools, methods, and technologies suitable for agriculture in local conditions.

While most of the sessions were delivered online, the participants also had a one-day tour of ICBA's head offices and research stations. During the tour, experts from ICBA and the Gracia Group arranged an interactive session for the students and their parents to discuss the importance of agriculture, the challenges that exist in the sector and the ways to contribute to sustainable agricultural development and food security in the UAE.

The students also visited the Emirates Soil Museum located at ICBA and made a virtual tour of the Gracia Farm. What is more, they got to plant native trees at ICBA as a way to make their contribution to environmental protection.

Ms. Ryma Affani, Curator of the Emirates Soil Museum, says: "We offer a wide range of learning programs for our visitors, but schoolchildren and students form the core part of our audience. Our goal is to educate them about the importance of sustainable agriculture through interactive activities."

The winter camp was held under ICBA's Youth Engagement Society, a program designed to facilitate communication and collaboration between young people in the UAE and their peers in other countries. The program's primary goal is to address local and global challenges related to food security and agricultural production in marginal environments and contribute to the achievement of the Sustainable Development Goals.

Harnessing digital technologies for capacity development

As digital technologies proved critical at the height of the pandemic, ICBA continued to expand its online capacity development programs in 2021.

The center continued equipping farmers, extension specialists, researchers, and young people, among others, with skills, resources, and tools to innovate and deal with various agricultural challenges. In particular ICBA launched a new series of online technical training courses to develop the knowledge of farming communities in Egypt and the UAE in sustainable agriculture in desert environments.

Dr. Charbel Tarraf, Chief Operations and Development, says: *"The pandemic presented an opportunity to create new programs and adapt existing ones for online learning. This, in turn, helped us to cater for a larger number of beneficiaries around the world."*

Held under the project titled "Food for the Future II - Building Sustainable Networks and Unleashing Entrepreneurial Potential of Farming Communities Living in Marginal Areas", the training series covered such topics as soil and water sampling; irrigation management; quinoa and Salicornia production; on-farm fertilizer, pesticide, and waste management; aquaculture; and marketing strategies in agriculture.

Funded by the HSBC Bank, the project's main goal is to provide farmers, agripreneurs, farm workers, and extension specialists with different tools and resources for sustainable agriculture under marginal conditions.

The project focuses on three components: climate-smart agricultural entrepreneurship in Egypt; unconventional sustainable farming approaches for salt-affected regions of Egypt; and capacity development in integrated farming in the UAE.

Organized from 10 February 2021 to 12 April 2021 in collaboration with Smart Labour, an educational platform, the courses were attended by about 500 participants from 49 countries, of whom 30 percent were women.



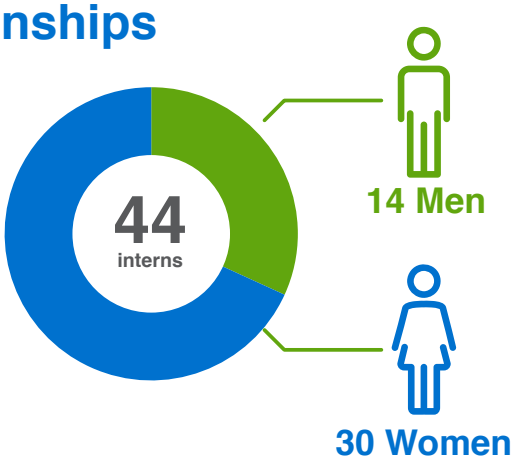
ICBA is diversifying the range and scope of its online educational offerings.

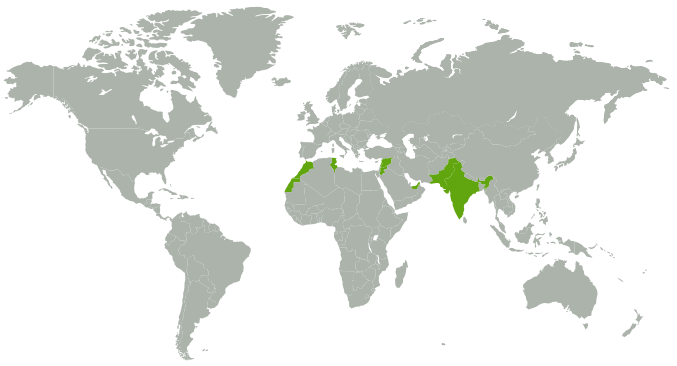
Developing capacities

Capacity development is an integral part of ICBA's work. In 2021 the center focused considerable efforts and resources on catering to the capacity development needs of different stakeholders.

ICBA continued to work on enhancing the capacities of institutions, researchers, students, and farmers through short- and medium-term training courses, workshops, farmer field schools, internships, master's, doctoral, and post-doctoral research programs.

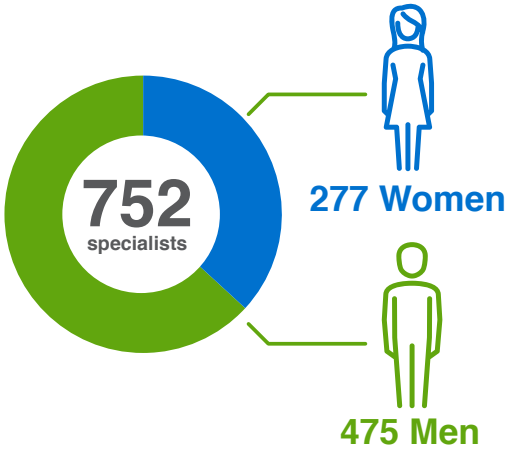
Internships

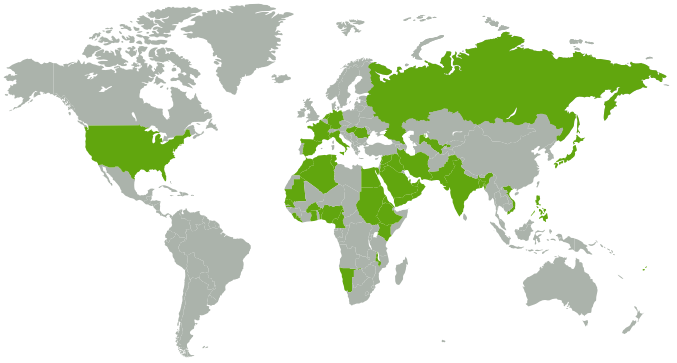




University students from nine countries

Technical training courses





16 special technical training courses, online and in-person, for participants from 51 countries

Communicating science-based knowledge



ICBA continued to communicate its science-based knowledge and research and development work to different audiences through the news media, its website, and social media channels.

267
media mentions

219,312
website sessions

336,605
YouTube views

Knowledge products



Developments in Soil Classification,
Land Use Planning and Policy Implications
HE UNITED ARAB EMIRATES
FAWZI M KARIM & ABDULLAH J DAKHEEL
WATER CONSERVATION STRATEGY
2010
Science, Policy and Politics of
Modern Agricultural System
United Arab Emirates Keys to Soil Taxonomy

Science-based publications form the core of ICBA's knowledge output. They are part of the center's contribution to the advancement of agricultural science. In 2021 our scientists produced a total of 74 research publications.

Peer-reviewed journals

Published

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Book chapters

Published

1. López-Martínez, A., Molina-Aiz, F. D., Valera, D. L. & **Fatnassi, H.** (2021). Geometric and Aerodynamic Characterization of Insect-Proof Screens Used in Greenhouses. Aerodynamic Models. In: Drs. Singh & Sharma (eds). Protected Cultivation, Modeling and Automation. Taylor & Francis Group.

2. Molina-Aiz, F. D., Valera, D. L., López-Martínez, A. & **Fatnassi, H.** (2021). Ventilation Models in Greenhouses. In: Drs. Singh & Sharma (eds). Protected Cultivation, Modeling and Automation. Taylor & Francis Group.

3. **Shahid, M.** & **Thushar, S.** (2021) Response of quinoa to high salinity under arid conditions. In: P. Vellinga et al. (eds.). Saline futures: state-of-the-art in saline agriculture, pp. 427-441. doi: 10.1201/9781003112327-29

4. Ranasinghe, R., Ruane, A. C., Vautard, R., Arnell, N., Coppola, E., Cruz, A. F., Dessai, S., Islam, A. S., Rahimi, M., Carrascal, D. R., Sillmann, J., Sylla, B. M., Tebaldi, C., Wang, W. & **Zaaboul, R.** (2021). Climate change information for regional impact and for risk assessment. IPCC Sixth Assessment Report (Working Group I)

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1. **Tutundjian, S.**, Clarke, M., Egal, F., Dixson-Decleve, S., Candotti, W. C., Schmitter, P. & Lovins, L. H. (2021). Future food systems: Challenges and consequences of the current food system. In: The Palgrave Handbook of Climate Resilient Societies. Springer.

2. **Kheir, A. M. S., Ammar, K., Attia, A.**, Elnashar, A., Ahmad, S., El-Gioushy, S. F. & Ahmed, M. (2022). Cereal crop modeling for food and nutrition security. In: Springer Book of Climate Change and Food Security.

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- The Institute of International Education
- The International Maize and Wheat Improvement Center
- University of Wollongong in Dubai
- UPL Management DMCC

Financials





Statement of financial position

	As at 31 December	
	2021	2020
	USD'000	USD'000
ASSETS	USD'000	USD'000
Non-current assets		
Property and equipment	3,684	3,839
Right-of-use asset	24	59
	3,708	**3,898**
Current assets		
Inventories	46	53
Account receivables	607	831
Short – term deposits	17,706	13,620
Cash and bank balances	3,961	7,238
	22,320	**21,742**
TOTAL ASSETS	**26,028**	**25,640**
EQUITY AND LIABILITIES		
Equity		
Reserves – Designated	15,397	15,397
Reserves – Undesignated	4,840	3,877
Total equity	**20,237**	**19,274**
LIABILITIES		
Non-current liabilities		
Defined benefit obligations	656	587
Deferred income – restricted funds	673	-
Lease liabilities	4	26
	1333	**613**
Current liabilities		
Trade and other payables	2,847	3,141
Deferred income – restricted funds	1,589	2,574
Lease liabilities	22	38
	4,458	**5,753**
Total liabilities	**5,791**	**6,366**
TOTAL EQUITY AND LIABILITIES	**26,028**	**25,640**

Statement of activities and other comprehensive income

	Year ended 31 December	
	2021	2020
	USD'000	USD'000
Grants income	9,730	8,999
Other income	22	21
Research and collaborator expenses	(7,255)	(6,004)
General and administration expenses	(1,698)	(2,169)
Operating (deficit)/surplus for the year	799	847
Finance income - net	164	268
Surplus for the year	963	1,115
Other comprehensive income	-	-
Total comprehensive income for the year	963	1,115

Board of directors

In 2021 ICBA welcomed a new board of directors appointed in line with a resolution by H.E. Mariam bint Mohammed Almheiri, Minister of Climate Change and Environment of the UAE. The board will serve a three-year term and comprises renowned leaders, scientists, and experts from governmental and non-governmental agencies, donor institutions, as well as research and development organizations.



H.E. Razan Khalifa Al Mubarak
Managing Director, Environment Agency - Abu Dhabi (Chair)



Mr. Mohammad Jamal Alsaati
Special Advisor, Office of the President of the Islamic Development Bank and Group Chairman



Ms. Aysha Al Suwaidi
Project Manager, Ministry of Climate Change and Environment of the UAE



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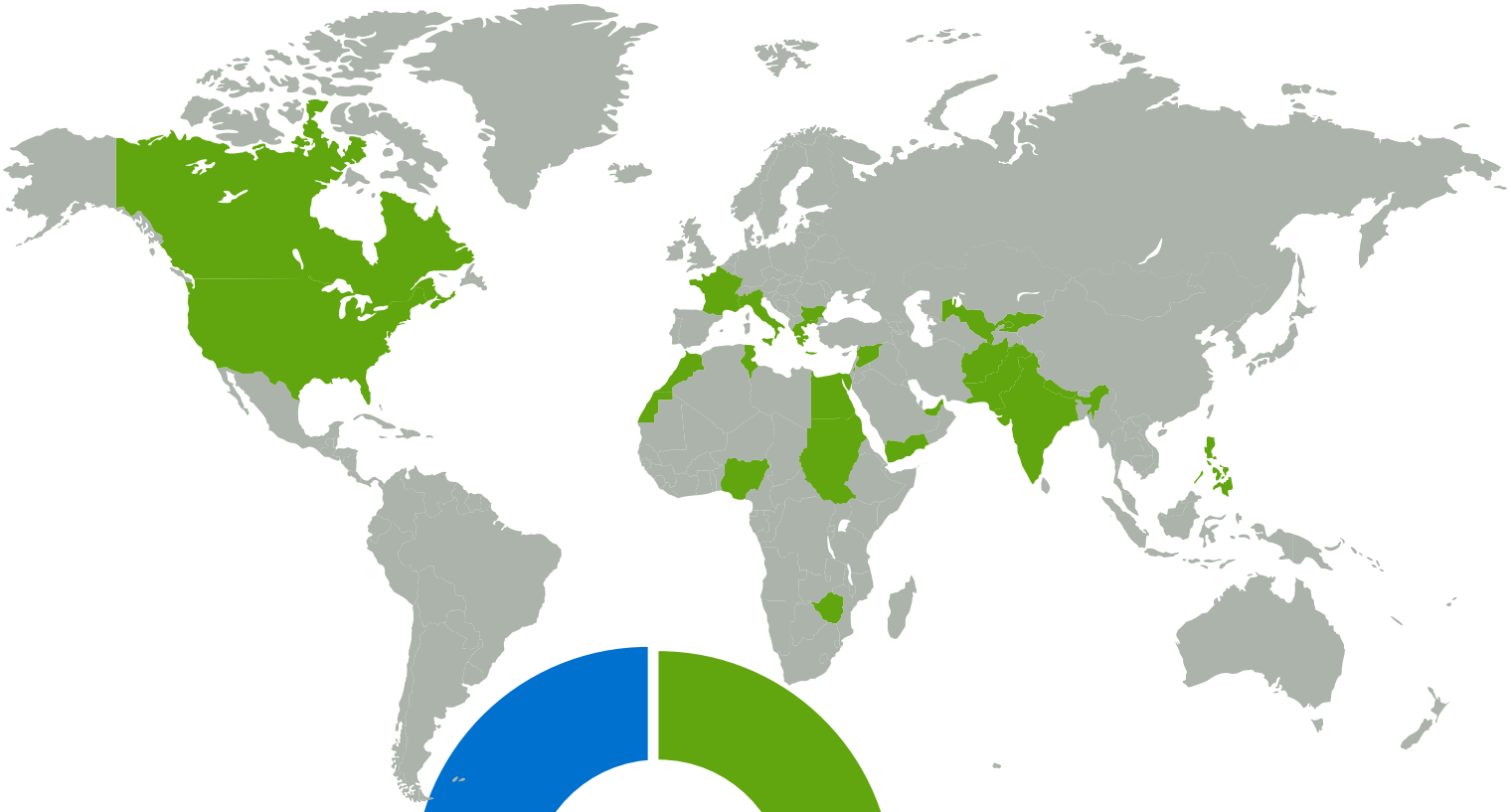


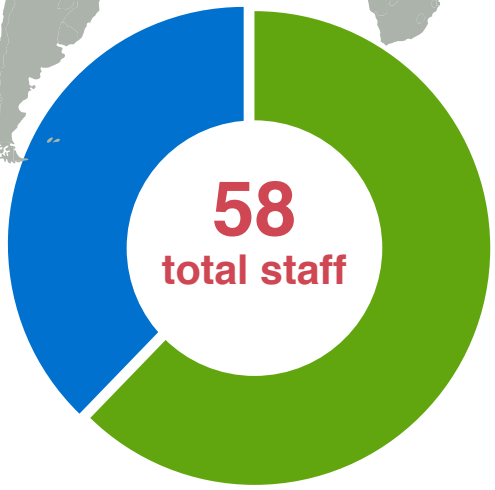
Dato' Dr. Mohamad Zabawi bin Abdul Ghani
Director General, Malaysian Agricultural Research and Development Institute



Dr. Tarifa Alzaabi
Director General, ICBA (Ex officio member)

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ICBA staff are from

23

countries around the world



17

women

41

men



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